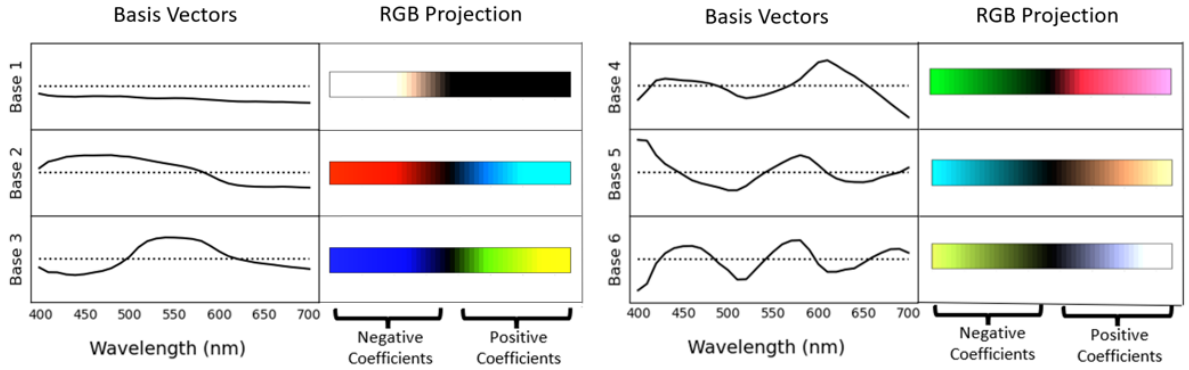
Reconstruction Algorithm

$$\min_{\mathbf{z}} \frac{1}{2}||\mathbf{y} - \mathbf{CHPz}||_2^2 + \Phi_{\boldsymbol{\theta}}(\mathbf{Pz})$$

Meas. Crop Conv matrix Reg. Dimension reduction

Basis P



Coupled with the forward model (HP and C).

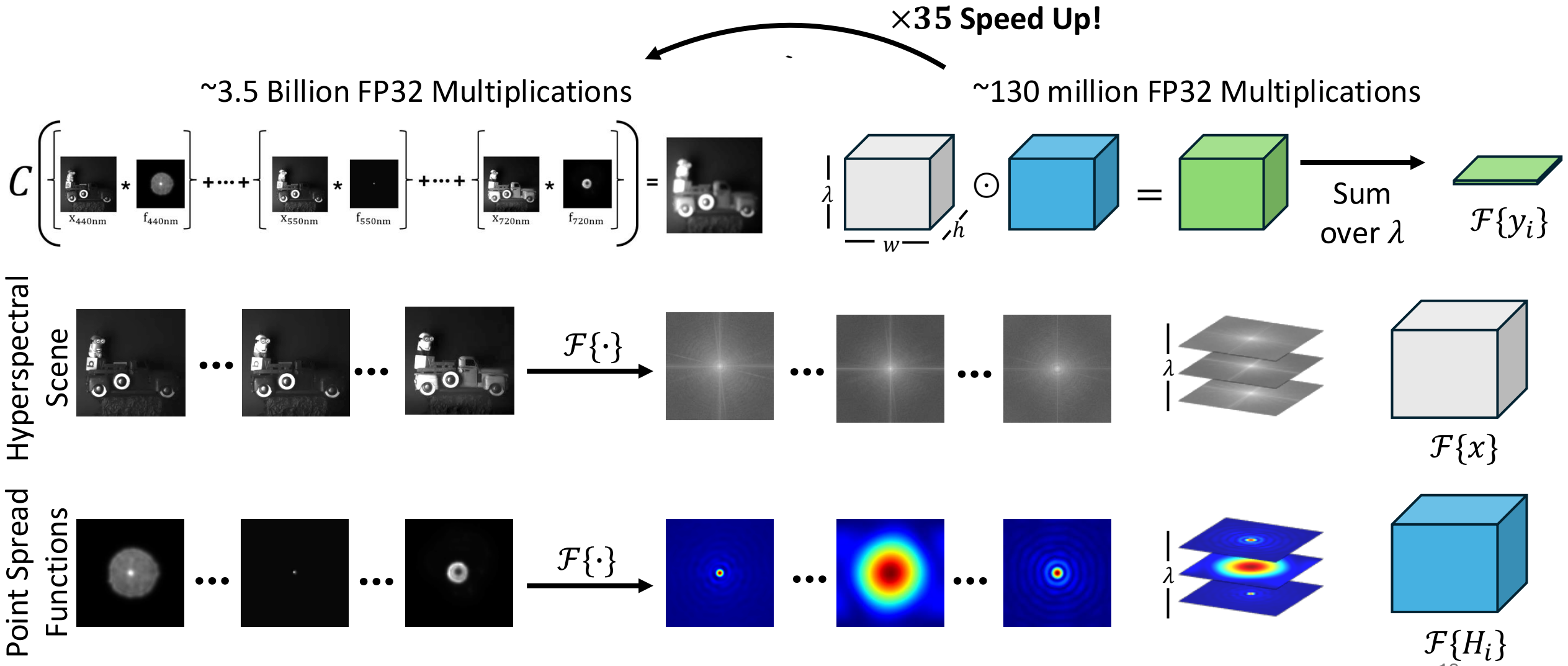
Off-the-shelf (no training required).

Physics-based Estimate

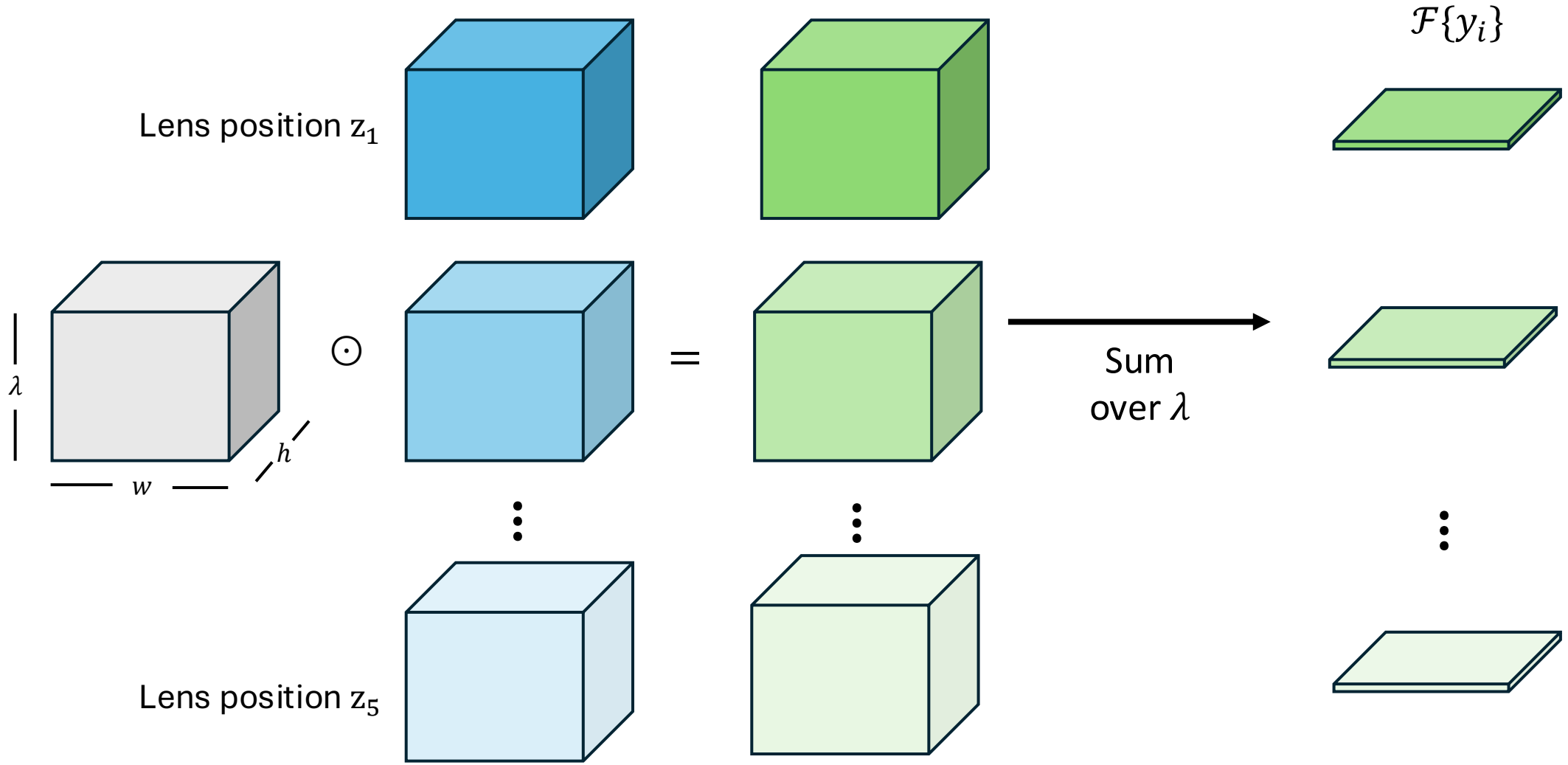
Deep-learning based Denoiser (Transformer)

Fast estimation requires careful design!

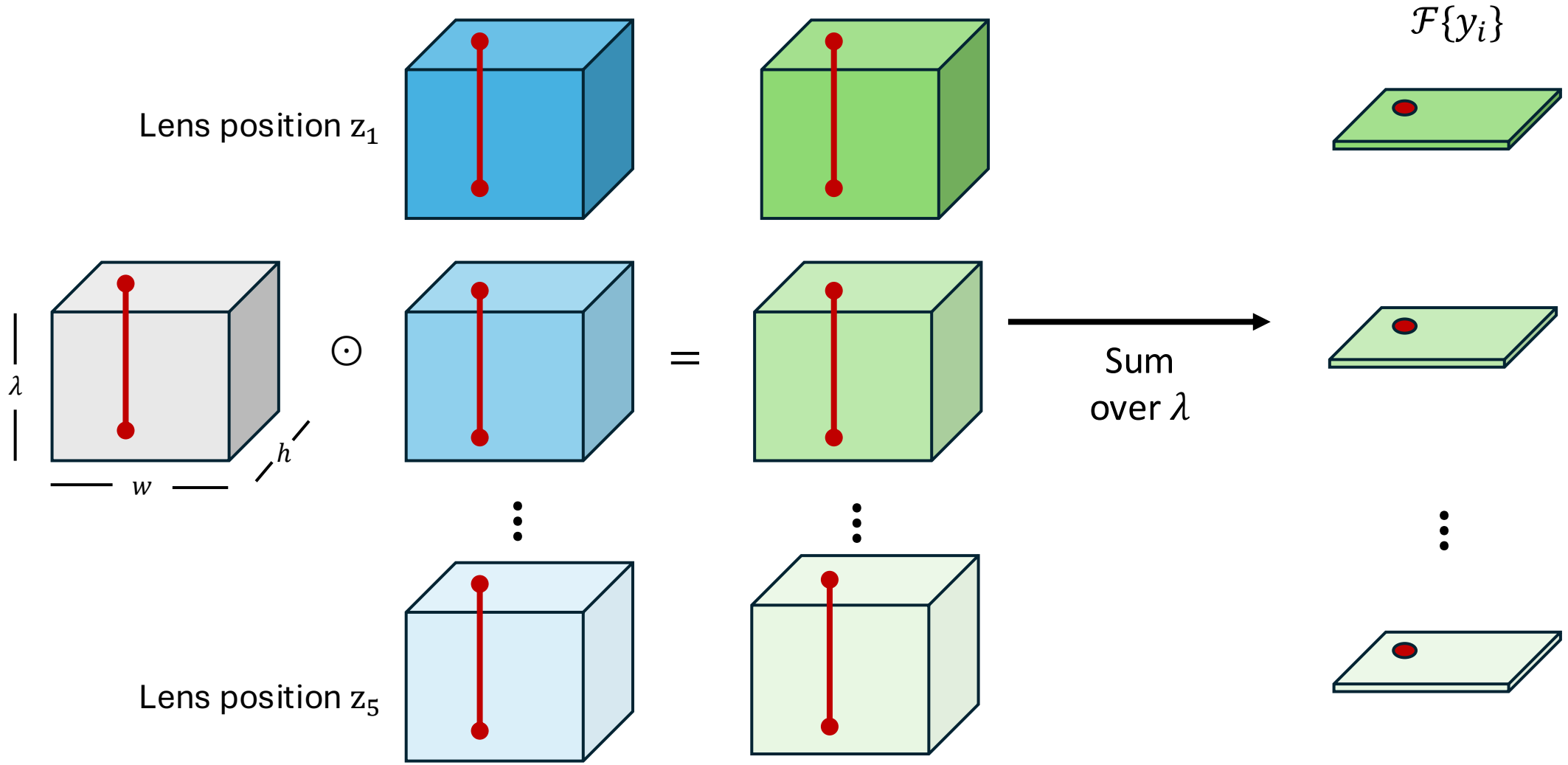
Defocus is Efficient in Computation



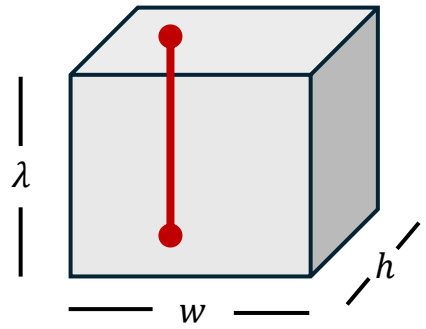
Defocus is Efficient in Computation

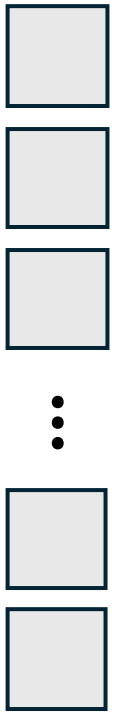


Defocus is Efficient in Computation

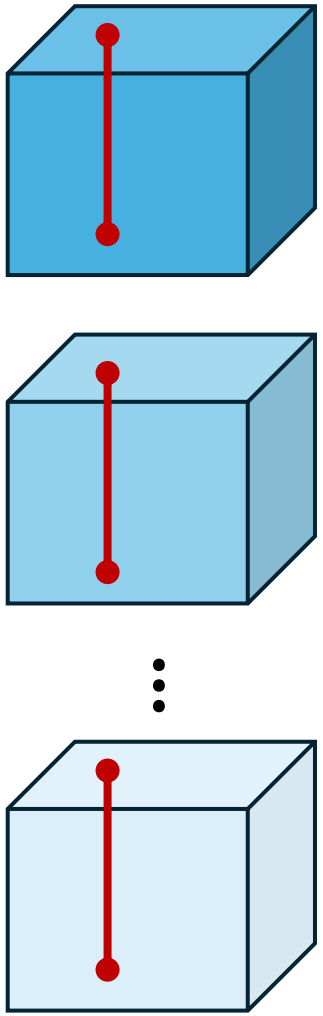


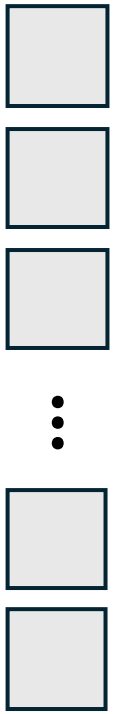
Defocus is Efficient in Computation



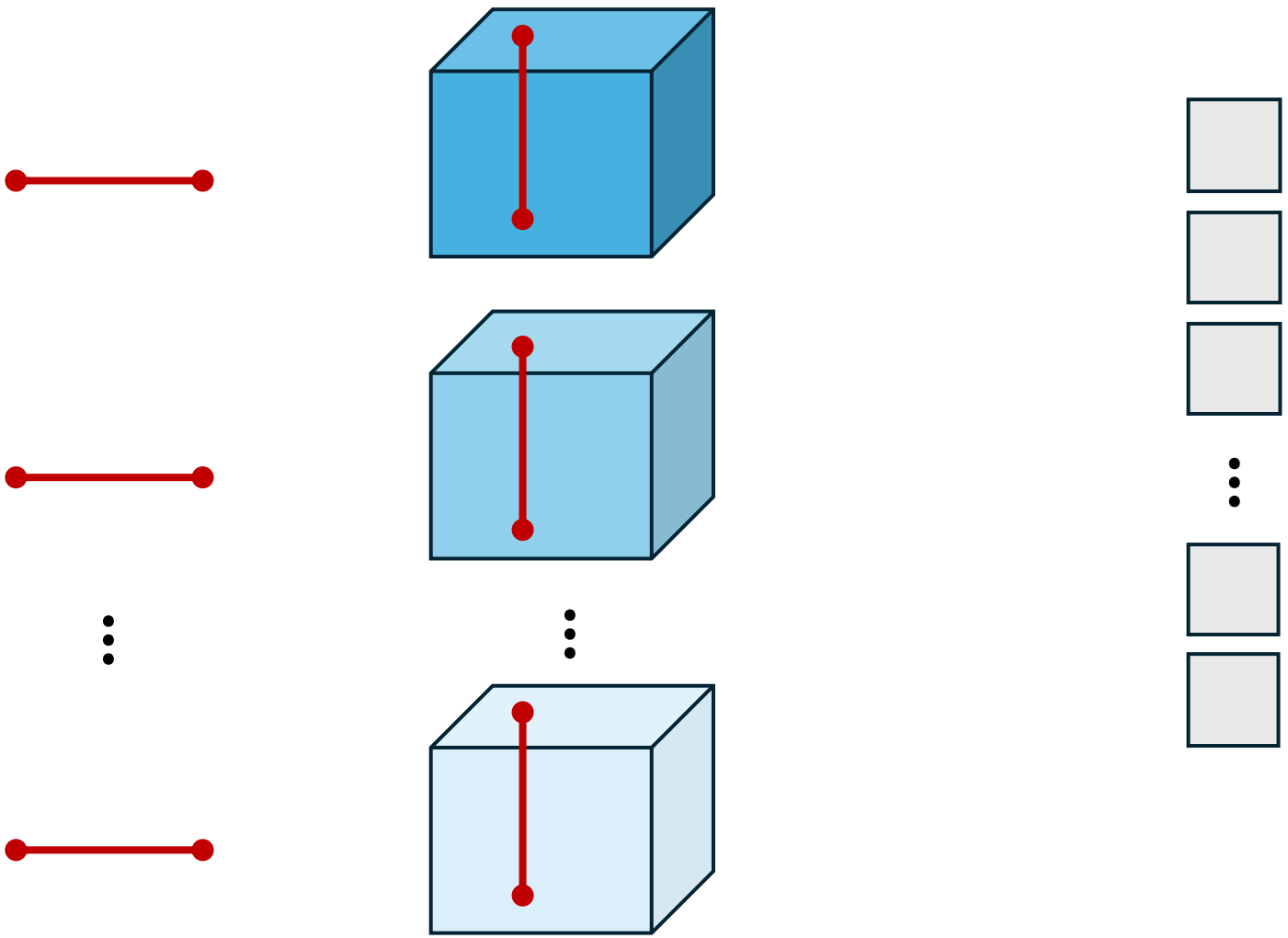


Defocus is Efficient in Computation

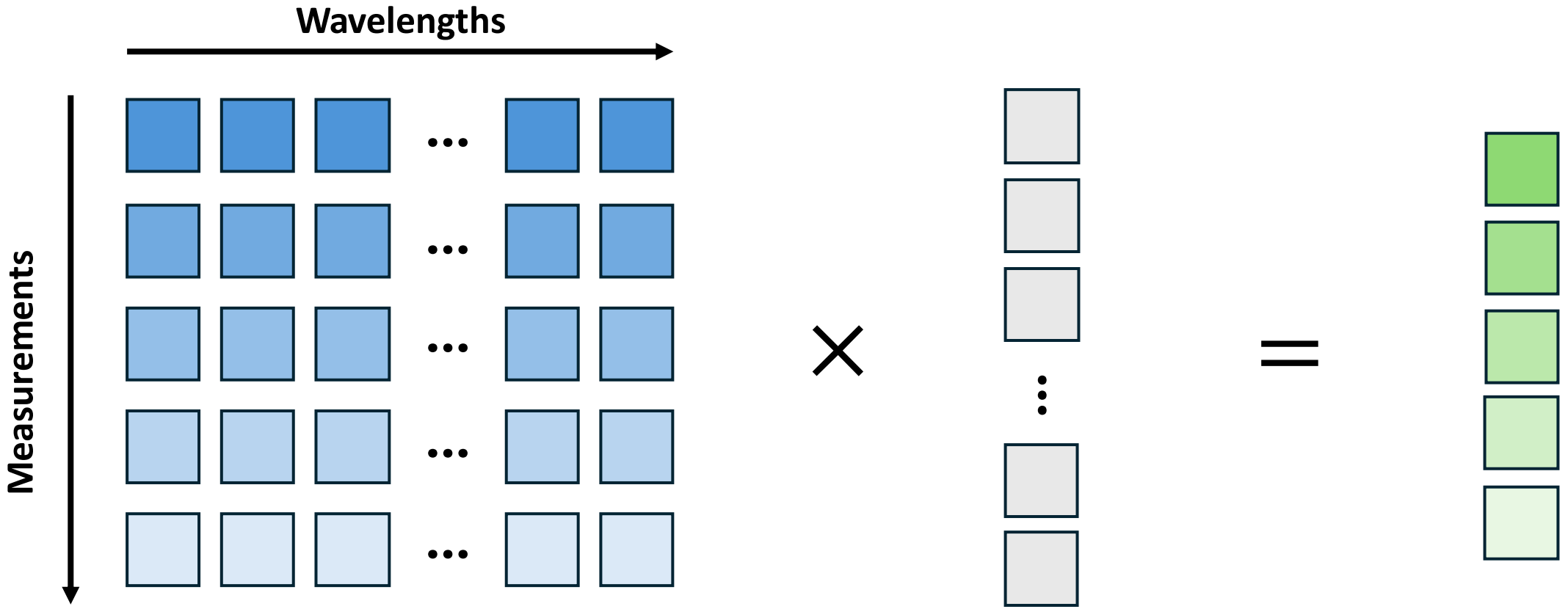




Defocus is Efficient in Computation

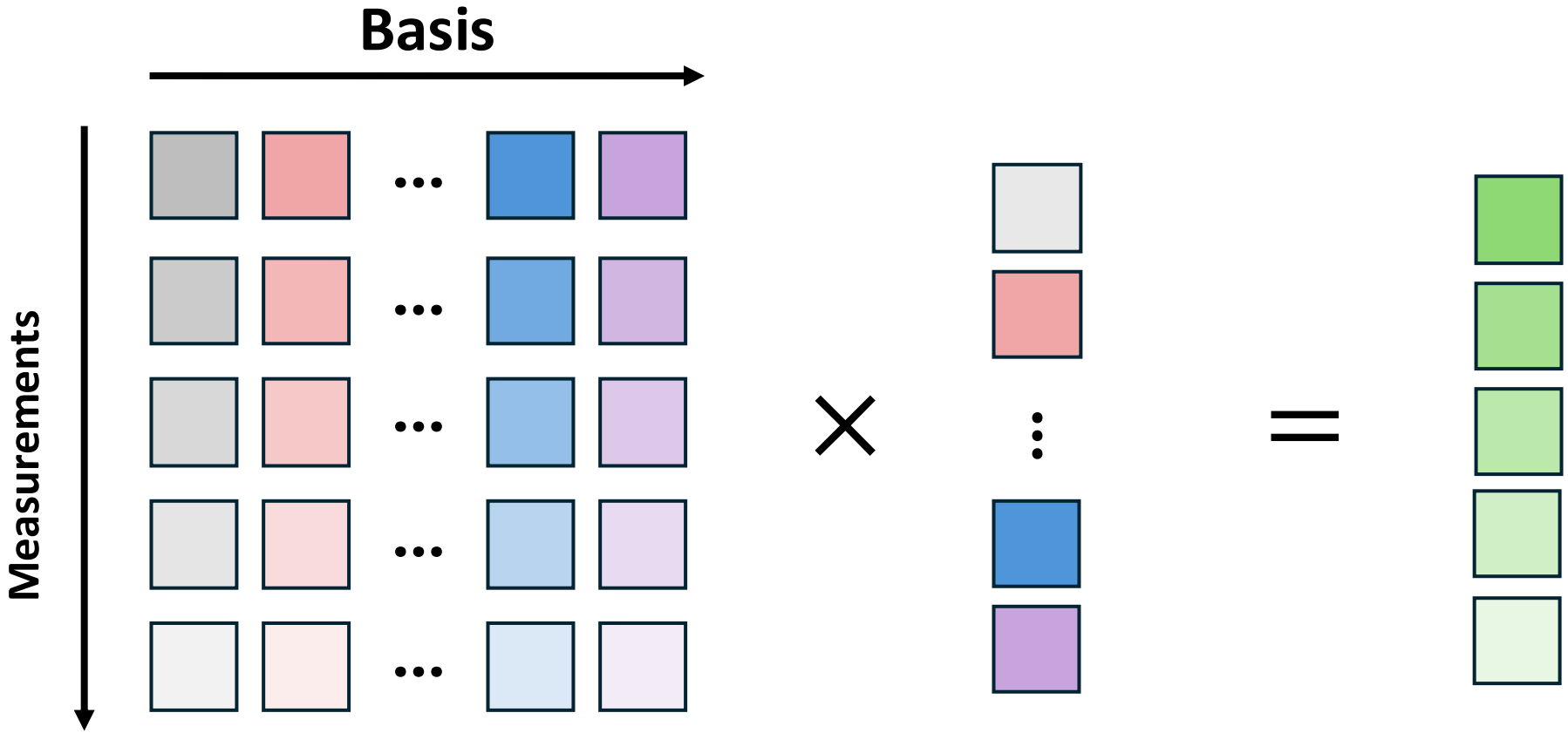


Defocus is Efficient in Computation



Each of the pixels forms an independent small linear system

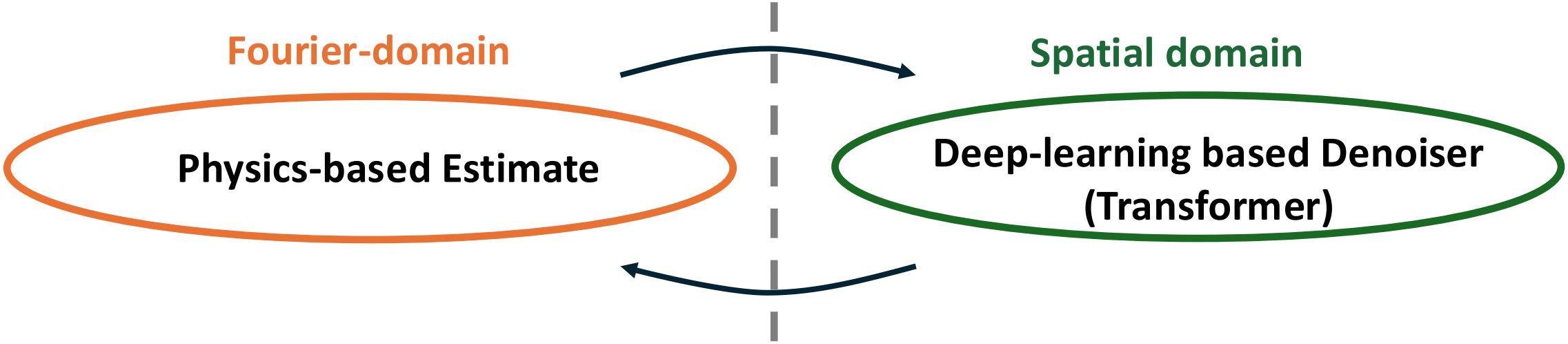
Defocus is Efficient in Computation



Changing basis doesn't break the structure, but decreases computation

Defocus is Efficient in Light and Compute

- **Convolution through Fourier-domain makes 35x speed up**
- **Computing through sub-systems enables better algorithms**



0.2 second per iteration
Converge in 3 iterations